

Lesson 06 Demo 02

Implementing Virtual Machine Availability Zone

Objective: To set up an availability set for virtual machines, ensuring high availability and reliability

Tools Required: Azure account

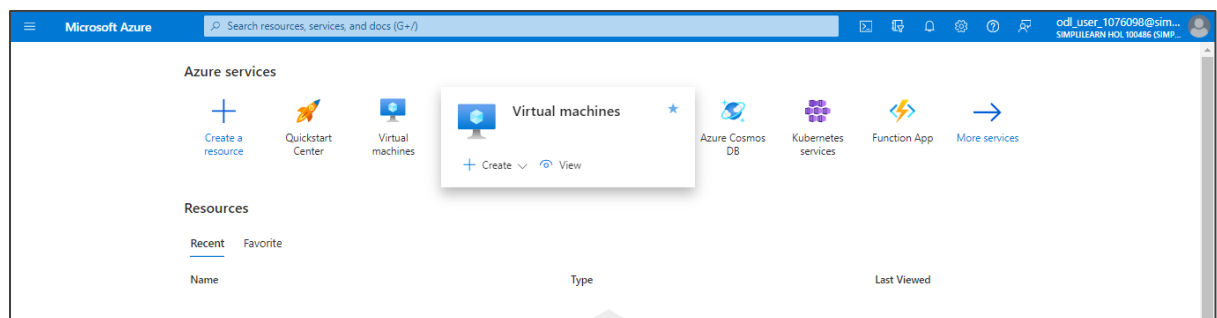
Prerequisites: None

Steps to be followed:

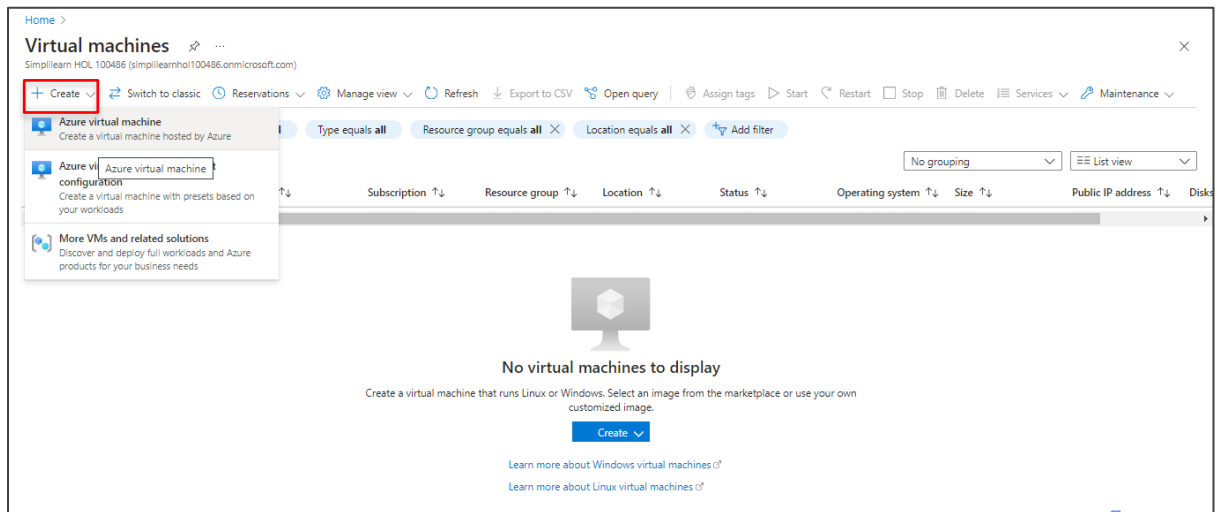
1. Configure the availability zone while creating a VM

Step 1: Configure the availability zone while creating a VM

1.1 Log in to the **Azure Portal** and search for **virtual machines**



1.2 Click on + Create and select Azure virtual machine



1.3 In the Basics tab, select the Subscription, create the Resource group, and enter the Virtual machine name. Now, select Availability zone under Availability options, and select Zone 1 under Availability zone.

The screenshot shows the 'Create a virtual machine' wizard in the 'Basics' tab. The wizard has several tabs: Basics, Disks, Networking, Management, Monitoring, Advanced, Tags, and Review + create. The 'Basics' tab is active, showing instructions to create a virtual machine that runs Linux or Windows. Below the instructions, there are two sections: 'Project details' and 'Instance details'. In the 'Project details' section, the 'Subscription' is set to 'Simplilearn HOL 100486' and the 'Resource group' is set to 'RG01'. In the 'Instance details' section, the 'Virtual machine name' is 'myvm1', the 'Region' is '(US) East US', the 'Availability options' are 'Availability zone', and the 'Availability zone' is 'Zone 1'.

1.4 Choose the **Security type** and **Image**, and enter the **Username** and **Password** for the **Administrator account**

1.5 Choose **HTTP (80)** and **RDP (3389)** under **Select inbound** ports, and click **Next: Disks >**

1.6 Select **Standard HDD (locally-redundant storage)** under **OS disk type** and click **Review + create**

Home > Virtual machines >

Create a virtual machine

Basics Disks Networking Management Monitoring Advanced Tags Review + create

Azure VMs have one operating system disk and a temporary disk for short-term storage. You can attach additional data disks. The size of the VM determines the type of storage you can use and the number of data disks allowed. [Learn more](#)

VM disk encryption

Azure disk storage encryption automatically encrypts your data stored on Azure managed disks (OS and data disks) at rest by default when persisting it to the cloud.

Encryption at host ☐

Encryption at host is not registered for the selected subscription. [Learn more about enabling this feature](#)

OS disk

OS disk type *

The selected VM size supports premium disks. We recommend Premium SSD for high IOPS workloads. Virtual machines with Premium SSD disks qualify for the 99.9% connectivity SLA.

Delete with VM ☒

Key management

[Review + create](#) < Previous Next: Networking > [Give feedback](#)

1.7 Click **Create**

Home > Virtual machines >

Create a virtual machine

Basics Disks Networking Management Monitoring Advanced Tags Review + create

Validation passed

Cost given below is an estimate and not the final price. Please use [Pricing calculator](#) for all your pricing needs.

Price

1 X Standard D2s v3 by Microsoft

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TERMS

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[Create](#) < Previous Next > [Download a template for automation](#) [Give feedback](#)

The final output appears as follows:

The screenshot shows the Azure portal interface for a deployment named 'CreateVm-MicrosoftWindowsServer.WindowsServer-202-20230921123729'. The deployment is in a 'Succeeded' state. The main content area displays 'Your deployment is complete' with details including the deployment name, subscription (Simplilearn HOL 100486), resource group (RG01), start time (9/21/2023, 12:42:39 PM), and correlation ID. It also lists recommended next steps: 'Setup auto-shutdown', 'Monitor VM health, performance and network dependencies', and 'Run a script inside the virtual machine'. A 'Go to resource' button is visible. On the right, a 'Deployment succeeded' notification is shown, along with sections for 'Cost Management' and 'Microsoft Defender for Cloud'.

The screenshot shows the 'Virtual machines' page in the Azure portal. It displays a table with 2 records. The table columns are: Name, Type, Subscription, Resource group, Location, Status, Operating system, Size, Public IP address, and Disks. The first record is 'myvm' and the second is 'myvm1'. Both are 'Virtual machine' type, located in 'East US', and are in a 'Running' state. They are both using 'Standard_D2s_v3' size and have a 'Public IP address'.

Name	Type	Subscription	Resource group	Location	Status	Operating system	Size	Public IP address	Disks
myvm	Virtual machine	Simplilearn HOL 100486	RG01	East US	Running	Windows	Standard_D2s_v3	172.190.137.67	1
myvm1	Virtual machine	Simplilearn HOL 100486	RG01	East US	Running	Windows	Standard_D2s_v3	172.171.251.113	1

By following the above steps, you have successfully set up an availability set for virtual machines to ensure high availability and reliability.